

Arab Academy for Science, Technology & Maritime Transport (AAST)



College of Computing & Information Technology

CSE3101: Software Requirements and Specifications

Dr. Mohamed Shaheen.

Eng. Salma Nagy.

Department: Software Engineering.

Requirements Engineering Artifacts Document.

PARROT BOOKS SALES SYSTEM

Ahmed Kotb.	232000349.
David Samir.	232000209.
Omar Essam.	241001994.



1. Requirement Categorization (10 Requirement Types)

Requirements extracted from the Parrot Books Sales System case study, classified using the ten requirement types and format. Every requirement below carries a unique ID (prefix + number) for traceability into the SRS and test plan.

Prefix	Requirement Type
BR	Business Requirement
BRULE	Business Rule
CON	Constraint
EIR	External Interface Requirement
FEA	Feature
FR	Functional Requirement (full numbered set FR1–FR31 in the SRS, Section 3)
NFR	Nonfunctional Requirement
QA	Quality Attribute
SR	System Requirement
UR	User Requirement

1.1. Business Requirements

Objective:

- BR1: Parrot Books wants to replace its manual, screen-based Sales Order and Sales Ledger processing with an integrated web-based Sales System.
- BR2: Speed up order-to-cash turnaround, eliminate re-keying of the same order data across departments, and give management real-time visibility into stock, orders, and receivables — supporting the firm's continued international growth while preserving its reputation for excellence and customer service.

1.2. Business Rules

Policies and Guidelines:

- BRULE1 (Order Validity Rule): a sales order is invalid, and must be referred back to the customer, if the firm does not supply the book(s) ordered or if the pricing details on the order are incorrect.
- BRULE2 (No Partial Shipment Policy): the firm only ships the complete ordered quantity for an item; part shipments are not made.
- BRULE3 (Publisher Grouping Rule): a separate purchase requisition is raised for books sourced from different publishers.
- BRULE4 (Statement Distribution Rule): a customer statement is produced only for customers who had a transaction in the last month or who have an outstanding balance.
- BRULE5 (Cheque-Only Payment Rule): cheque is the only payment method the system accepts from customers; cash, credit card, and bank transfer are not supported in Release 1.0.

1.3. Constraints

Restrictions Imposed:

- CON1 (Technology Constraint): the application must be a web-based system, accessible via standard browsers with no client-side installation.
- CON2 (Currency Constraint): all monetary values must be represented and displayed in Egyptian Pounds (EGP), matching the company's existing documents.
- CON3 (Data Integrity Constraint): a dispatch note cannot reference quantities greater than those confirmed on its picking ticket, and no order may be invoiced more than once.
- CON4 (Design Constraint): screen layouts should mirror the field layout of the existing paper documents (order, acknowledgement, picking ticket, dispatch note, invoice, remittance advice) to minimize staff retraining.

1.4. External Interface Requirements

Connections Between Systems and Users:

- EIR1 (Data Input from Publishers): Goods Received Note data must be entered by warehouse staff once publishers deliver against a purchase order (manual channel in Release 1.0).
- EIR2 (Output Interface to the Nominal Ledger): invoice and bank-deposit data must be provided to update the company's existing nominal/general ledger.
- EIR3 (Customer-Facing Interface): a screen must allow customers to place orders and view acknowledgements, invoices, and statements.
- EIR4 (Cashier/Bank Interface): the system must prepare bank deposit slip data for the cashier; physical bank submission remains manual.

1.5. Features

System Capabilities:

- FEA1 (Sales Order Management): capture, validate, and acknowledge customer orders.
- FEA2 (Stock and Purchasing): check stock on hand and generate purchase requisitions grouped by publisher.
- FEA3 (Warehouse and Fulfillment): generate picking tickets and dispatch notes.
- FEA4 (Sales Ledger and Invoicing): generate invoices and monthly customer statements.
- FEA5 (Payments and Cash Receipts): match remittances to invoices and prepare bank deposits.
- FEA6 (Reporting and Administration): maintain catalogue/customer data and produce operational reports.

1.6. Functional Requirements

System Behaviors Under Specific Conditions:

- FR1: If an ordered book is not in the catalogue or its price does not match, the system shall reject the line and return the order to the customer.
- FR2: If Quantity on Hand is insufficient for an order line, the system shall automatically generate a purchase requisition grouped by publisher.
- FR3: Once stock is confirmed, the system shall generate the sales order acknowledgement and calculate the extended price and order total automatically.
- FR4: Once a dispatch note is created, the system shall auto-draft the sales invoice from the dispatch, order, and pricing data with no manual re-entry.
- The complete, numbered set of functional requirements per feature (FR1–FR31) is in the SRS, Section 3.

1.7. Nonfunctional Requirements

System Properties/Constraints:

- NFR1 (Usability): the system must be easy to use for sales, warehouse, and accounts staff without extensive retraining.
- NFR2 (Performance): common screens (order entry, invoicing) must respond within 2 seconds even under normal daily order volume.
- NFR3 (Data Integrity): the system must ensure consistent stock, ledger, and order-status updates, with no duplicate invoicing or double stock deduction.

1.8. Quality Attributes

Service or Performance Characteristics:

- QA1 (Reliability): the system must reliably generate accurate acknowledgements, picking tickets, despatch notes, invoices, and statements every time, matching the case study's sample documents.
- QA2 (Scalability): the system must handle a growing number of orders, customers, and publishers as the company's international business grows.
- QA3 (Availability): the system must be accessible at least 99.5% of scheduled operating hours.

1.9. System Requirements

Top-Level Requirements:

- SR1: The system must integrate the previously separate, manual Sales Order Processing and Sales Ledger functions into a single web-based platform, sharing one order record end to end.
- SR2: The system must support role-based access for sales, warehouse, shipping, credit control, cashier, and administrator user classes, each with access limited to their function.
- SR3: The system must operate on standard workstation/browser combinations with no specialized hardware, while remaining reachable by customers internationally.

1.10. User Requirements

User Tasks and Goals:

- UR1 (Sales Staff): need to quickly capture and validate a customer order and generate an acknowledgement without re-entering data on separate screens.
- UR2 (Warehouse and Shipping Staff): need accurate, up-to-date stock information and a picking ticket/dispatch note generated directly from the order.
- UR3 (Credit Control / Accounts Staff): need invoices and statements generated automatically from dispatch data, and a reliable way to match customer payments to invoices.
- UR4 (Customers): need to place orders and receive timely, accurate acknowledgements, invoices, and statements reflecting the correct items, quantities, and prices.

2. Stakeholder Identification

2.1. Primary Stakeholders

Stakeholder	Role & Responsibilities	Expectations	Interaction with the System
Sales Staff	Receive, validate, and enter customer orders; prepare acknowledgements	Fast, accurate order entry with minimal re-keying	Direct daily users of order entry, validation, and acknowledgement screens
Warehouse & Shipping Staff	Pick items and prepare dispatch notes	Clear, accurate picking tickets; no ambiguity on availability	Direct daily users of picking ticket and dispatch note screens
Credit Control / Accounts Staff	Raise invoices, statements; match payments	Automatic, error-free invoicing and reliable audit trail	Direct daily/monthly users of invoicing, statement, and payment screens
Customers	Place orders; receive acknowledgements, invoices, statements	Timely, accurate order fulfillment and billing	Indirect (Release 1.0 order placement) / recipients of generated documents
Company Executives / Management	Set business direction; fund and sponsor the project	Measurable ROI, faster cycle times, real-time visibility	Consumers of reports and dashboards; approve scope changes
Purchasing / Buying Staff	Review and approve purchase requisitions; place purchase orders with publishers	Clear shortfall data grouped by publisher; minimal manual re-entry	Direct daily users of the requisition/purchase order screens
Cashier	Matches incoming cheques and prepares the daily bank deposit	Accurate, ready-to-submit deposit slip data	Direct daily user of the payment/deposit screen

2.2. Secondary Stakeholders

Stakeholder	Role & Responsibilities	Expectations	Interaction with the System
Publishers	Supply books in response to purchase orders	Clear, correctly grouped purchase orders	Indirect — receive system-generated purchase order data via existing manual channel
Bank	Processes the firm's deposits	Correctly totaled deposit slips	Indirect — receives physical deposit prepared from system data
IT/System Administrator	Maintains users, catalogue, and system configuration	Reliable, secure, maintainable system	Direct administrative user of the Reporting & Administration feature
Course Instructor / TA (project context)	Evaluates the requirements engineering deliverables	Complete, consistent, well-modelled documentation	Reviews the SRS, Vision & Scope, and this artifacts document
Auditors / Finance Department	Periodically review sales, invoicing, and payment records	Complete audit trail of orders, invoices, and payments	Indirect — consumes reports and exported ledger data
Marketing Department	Uses sales and customer data for promotions and reporting	Accurate, exportable sales history by customer/title	Indirect — future consumer of Reporting & Administration exports
Delivery / Carrier Partners	Physically transport dispatched orders to customers	Accurate dispatch note details (address, contents)	Indirect — receive dispatch note information printed from the system

3. Context Diagram (Level 0 DFD)

The diagram below treats the Parrot Books Sales System as a single process (Process 0) and shows the external entities it exchanges data with. Warehouse/Shipping and Credit Control are modelled as external departments relative to the core order-to-cash process for the purpose of this Level 0 view; their internal activities are decomposed in the Feature Tree and Use Case Diagram.

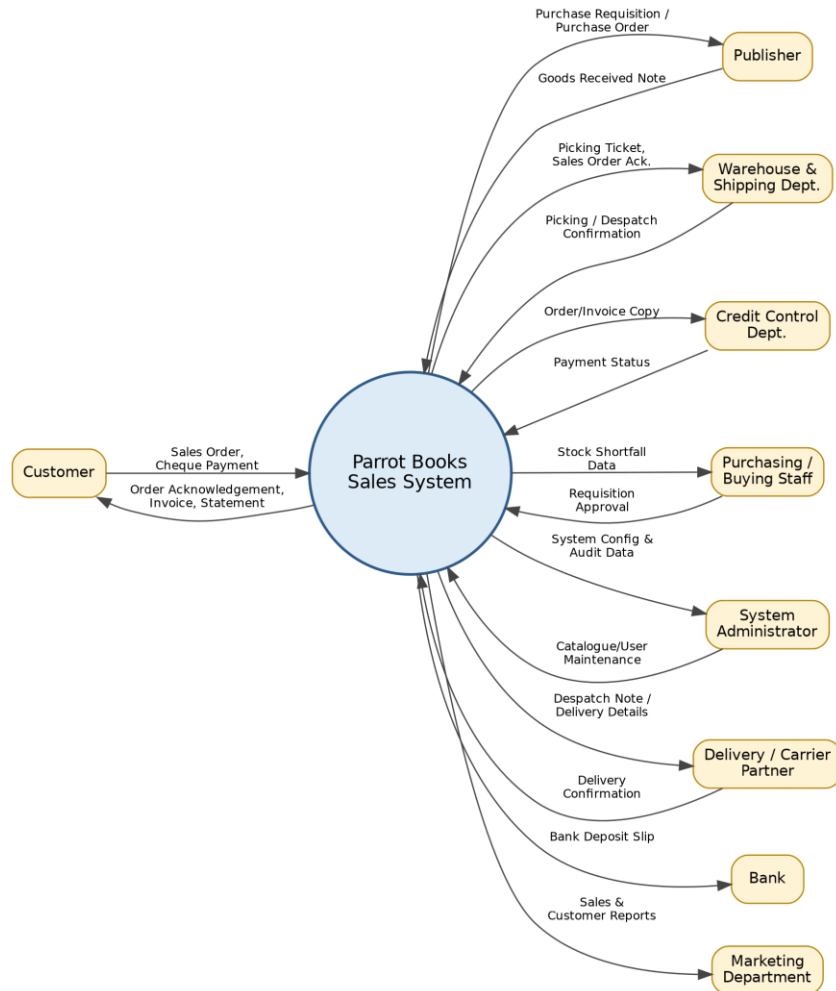


Figure 1 — Level 0 Context Diagram

4. Feature Tree

The feature tree decomposes the system into six high-level features, each with sub-features tracing directly to the functional requirements in the SRS (Section 3).

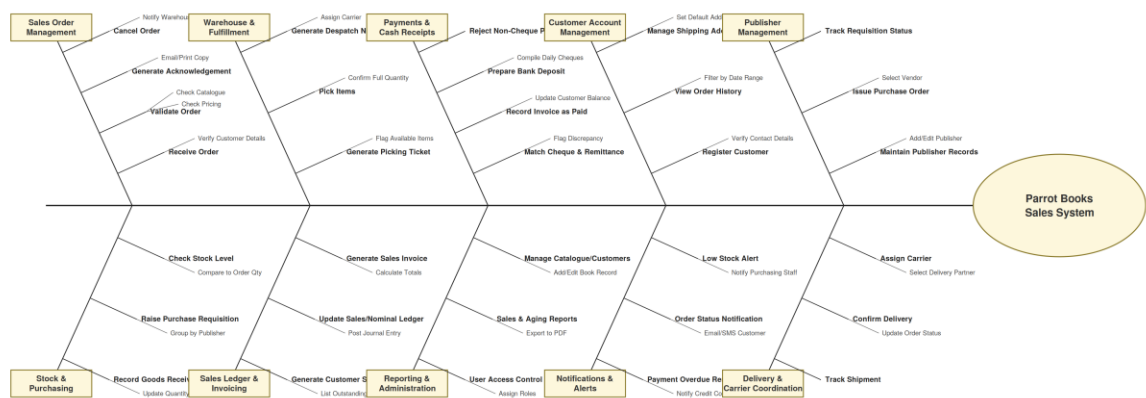


Figure 2 — Feature Tree

5. Use Case Diagram & Usage Scenario

5.1. Use Case Diagram

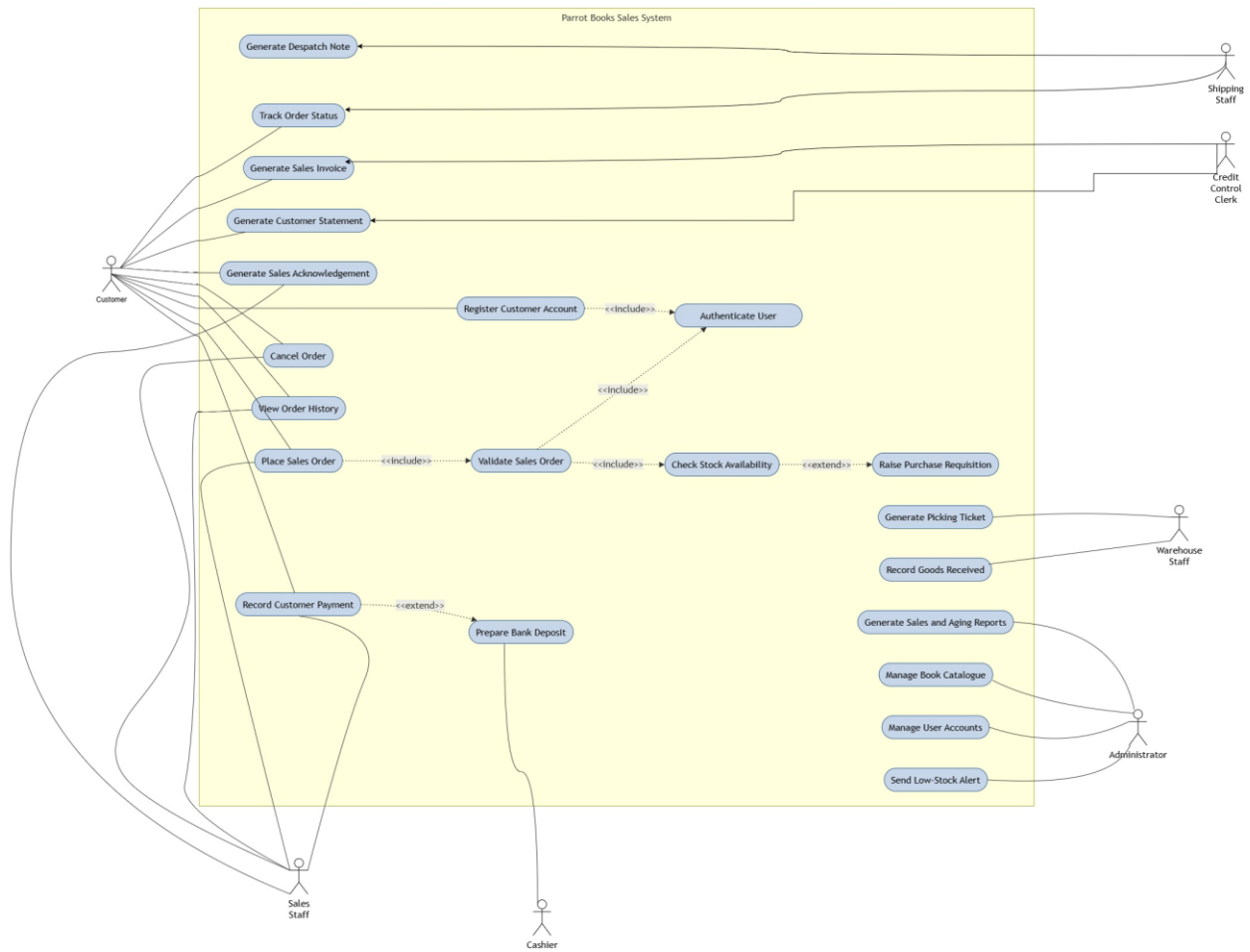


Figure 3 — Use Case Diagram (staff actors on the left, Customer on the right)

5.2. Usage Scenario — Fully Dressed Use Case

ID and Name:	UC-1 Request a Sales Order
Created By:	Team O.D.A. Date Created: 01/08/2026
Primary Actor:	Customer Secondary Actors: Sales Staff, Warehouse Staff, System
Description:	The Customer specifies the desired books to order, either by browsing the catalogue or entering ISBNs directly, and how many of each. The system validates the order against the book catalogue and current pricing, checks Quantity on Hand for each line, and either generates an acknowledgement or raises a purchase requisition for any shortfall items.
Trigger:	Customer indicates that they want to place a sales order.
Preconditions:	PRE-1. Customer's identity/account has been verified. PRE-2. Book catalogue and pricing data are current. PRE-3. Sales order system is online.
Postconditions:	POST-1. Order is stored in the Sales Order system. POST-2. Acknowledgement was sent to the Customer, or the order was rejected with a reason.
Normal Flow:	1.0 Place and Acknowledge a Sales Order 1. Customer specifies the desired books and quantities. 2. System validates each line against the book catalogue and current pricing. 3. System checks Quantity on Hand for each line (see 1.1). 4. System calculates the extended price and order total. 5. System generates the sales order acknowledgement and stores the order. 6. System notifies the Customer and the Warehouse.
Alternative Flows:	1.1 Order Requires a Purchase Requisition 1. At step 3, System determines that Quantity on Hand is insufficient for one or more lines. 2. System raises a purchase requisition for the shortfall items, grouped by publisher. 3. System sets order status to Awaiting Stock and notifies the Customer. 4. Normal flow resumes at step 5 once stock is replenished (see UC-2 Record Goods Received).
Exceptions:	1.2.E1 Book Is Not Supplied or Pricing Is Incorrect 1. System displays message: item not supplied, or price does not match the catalogue. 2. System asks Customer if they want to revise the order (2a) or cancel (2b). 2a. Customer revises the order; system restarts normal flow at step 2. 2b. Customer cancels the order; system terminates the use case.
Priority:	High
Frequency of Use:	Approximately 30–50 times per day across all sales staff and customer self-service combined.
Business Rules:	BRULE1, BRULE2, BRULE3
Other Information:	The system must calculate the extended price and order total automatically, and generate the acknowledgement with no manual re-entry of order data.
Assumptions:	Customer account and catalogue data are assumed to be accurate and current at the time of order placement.

6. Class Diagram

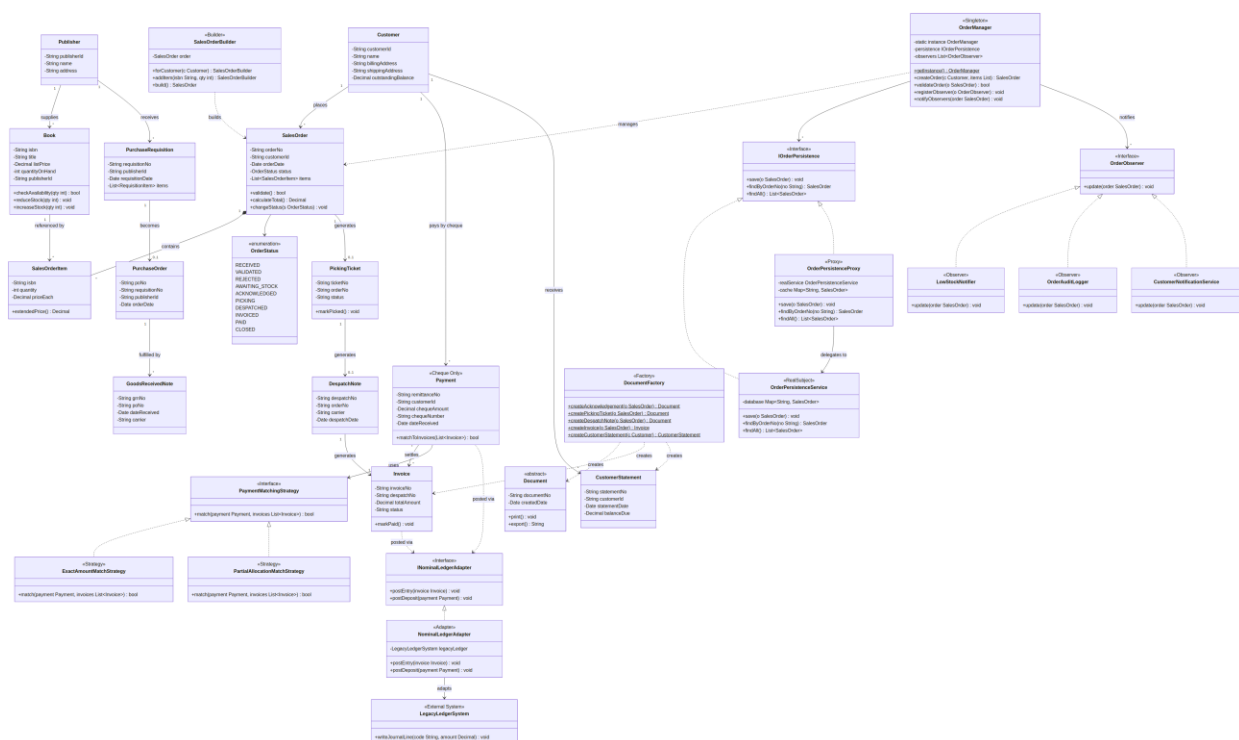


Figure 4 — Class Diagram (entities plus Singleton, Proxy, Observer, Factory, Strategy, Builder, and Adapter patterns)

7. Activity Diagram (Swimlanes Diagram)

The activity diagram uses swimlanes to show which steps are performed by the Customer versus the System when a sales order is placed and validated, complementing the Usage Scenario in Section 5.2.

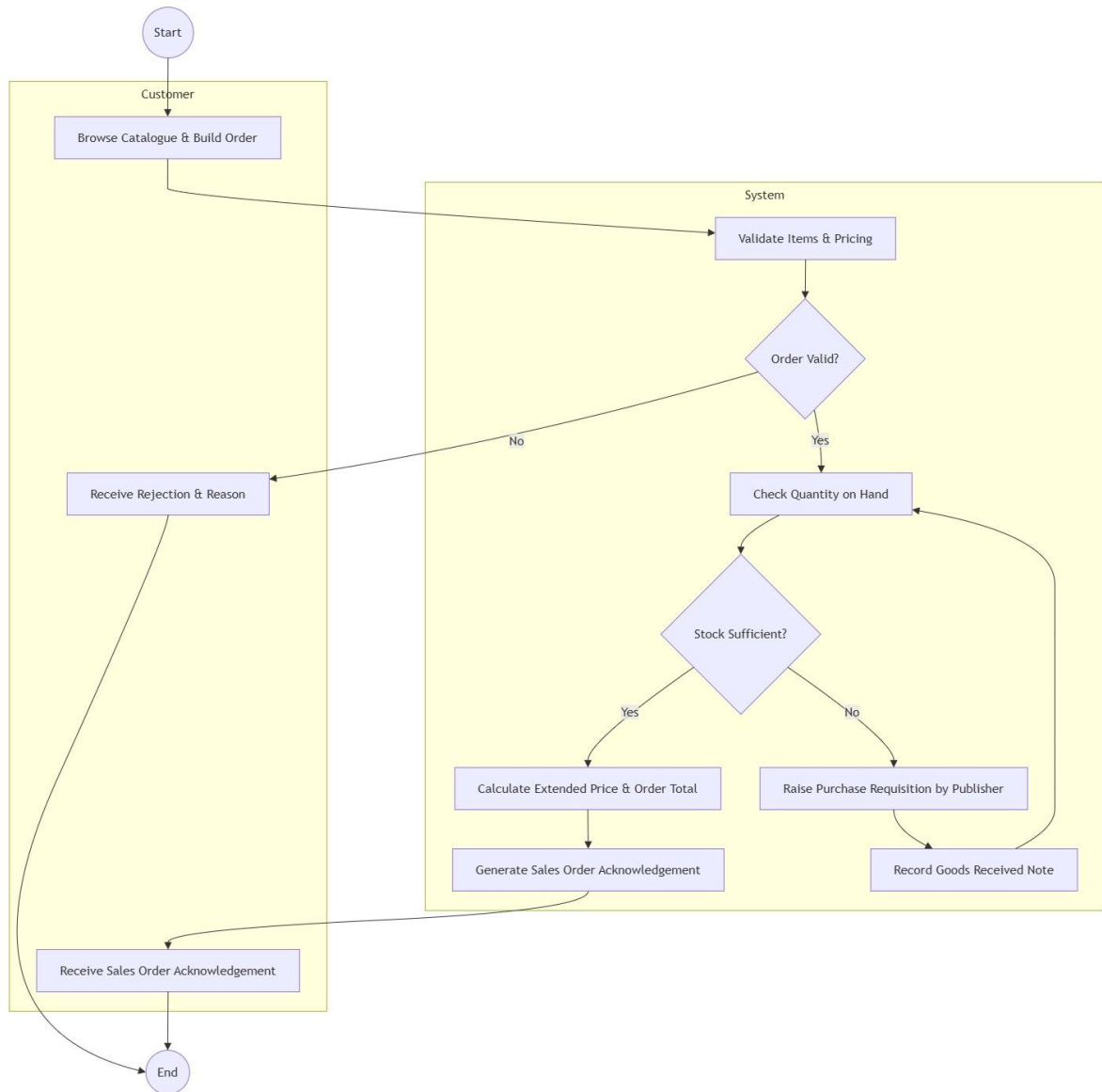


Figure 5 — Activity Diagram: Place and Validate Sales Order (Customer | System swimlanes)

8. State Chart Diagram & State Table

8.1. State Chart Diagram — Sales Order Lifecycle

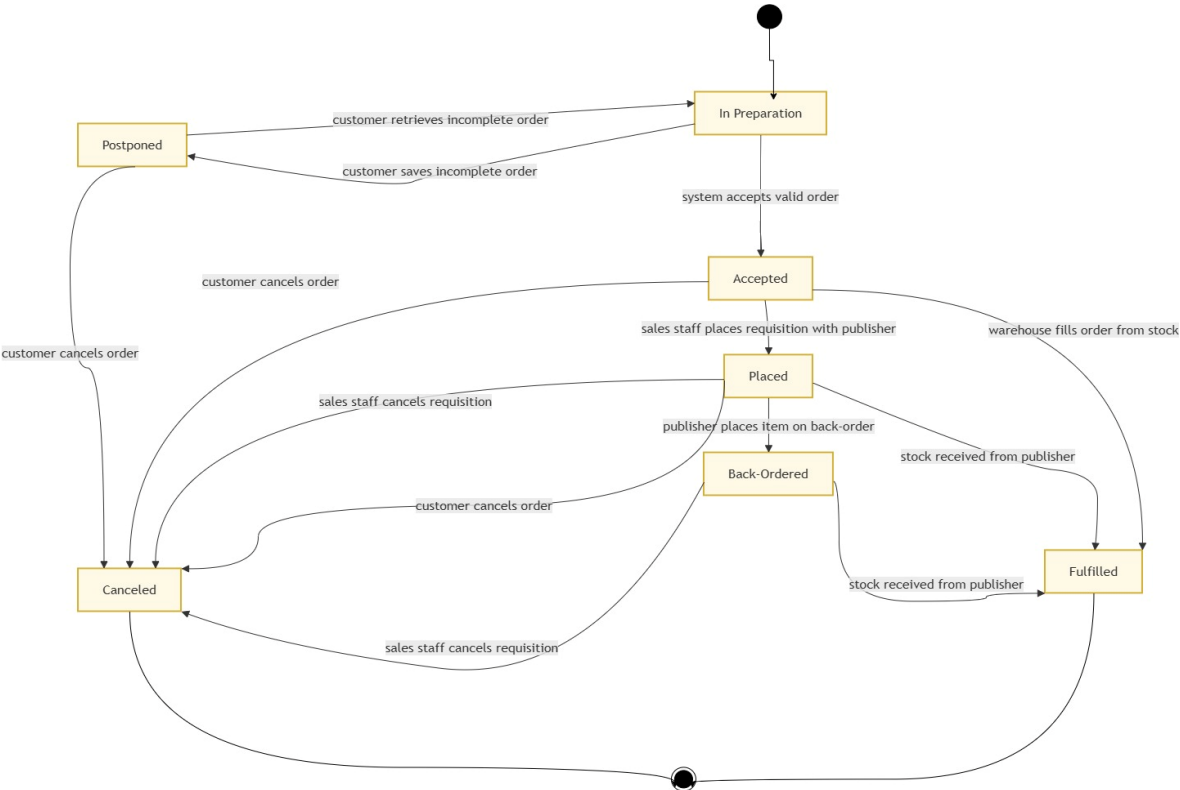


Figure 6 — State Chart Diagram for the Sales Order entity

8.2. State Table

	In Preparation	Postponed	Accepted	Placed	Back-Ordered	Fulfilled	Canceled
In Preparation	no	customer saves incomplete order	system accepts valid order	no	no	no	no
Postponed	customer retrieves incomplete order	no	no	no	no	no	customer cancels order
Accepted	no	no	no	sales staff places requisition with publisher	no	warehouse fills order from stock	customer cancels order
Placed	no	no	no	no	publisher places item on back-order	stock received from publisher	sales staff cancels requisition
Back-Ordered	no	no	no	no	no	stock received from publisher	sales staff cancels requisition
Fulfilled	no	no	no	no	no	no	no
Canceled	no	no	no	no	no	no	no

9. Sequence Diagram

The sequence diagram shows the message flow between the Customer, the System, and the Database when a sales order is submitted, including the alternative paths for an invalid order and for insufficient stock.

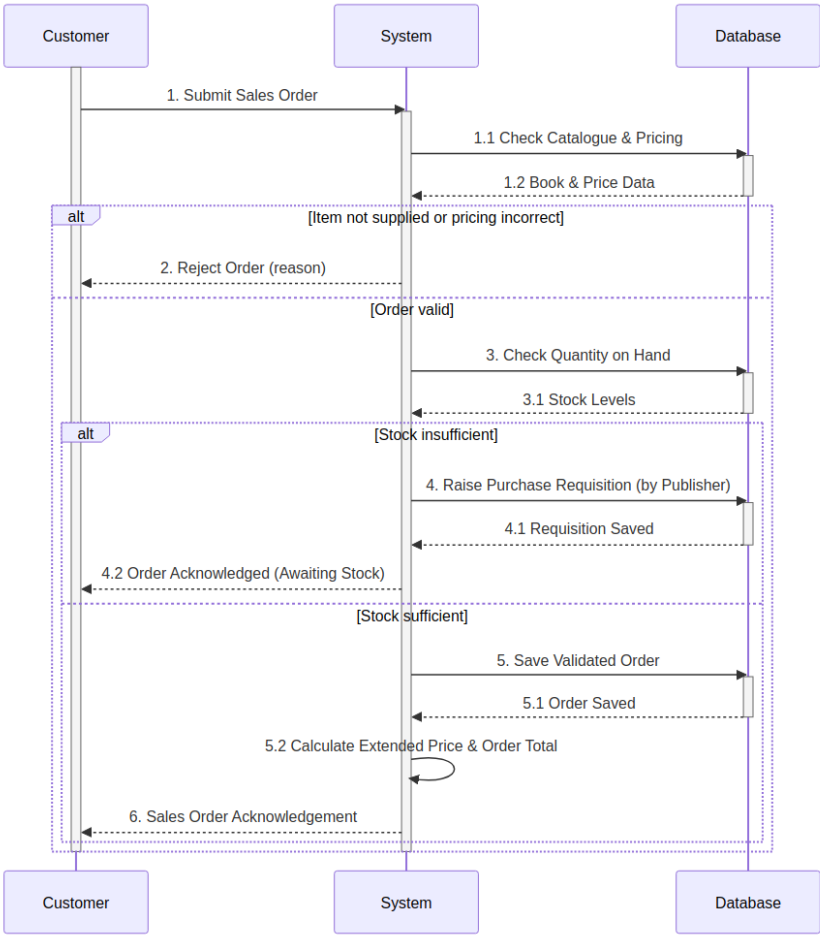


Figure 7 — Sequence Diagram: Submit and Process Sales Order

10. Dialogue Map

The dialogue map shows screen-to-screen navigation for the new web-based system, supporting the UI/UX design referenced in the SRS (Section 4.1).

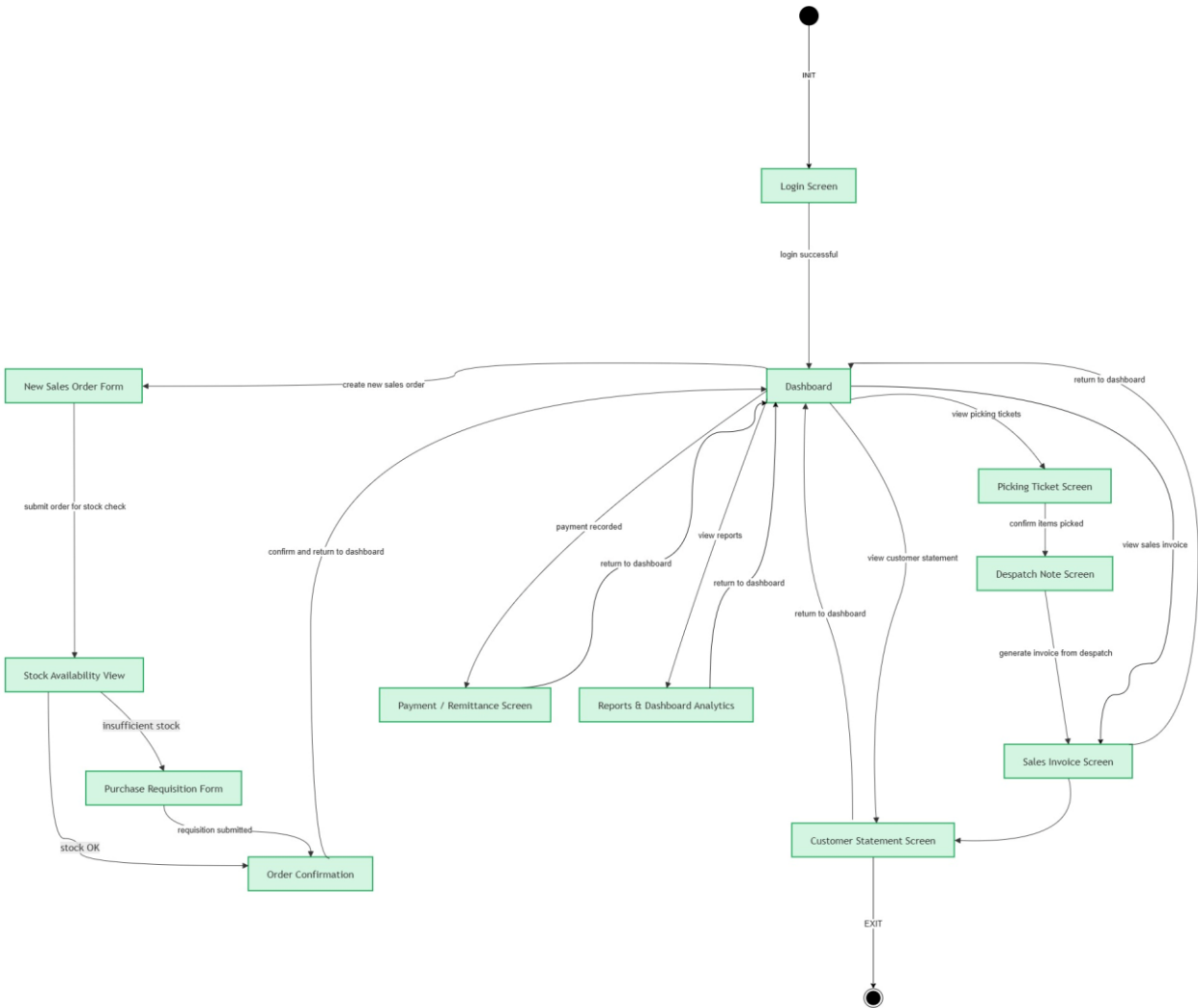


Figure 8 — Dialogue Map (screen navigation flow)

11. Decision Tree & Decision Table

11.1. Decision Tree — Order Validation & Stock Decision

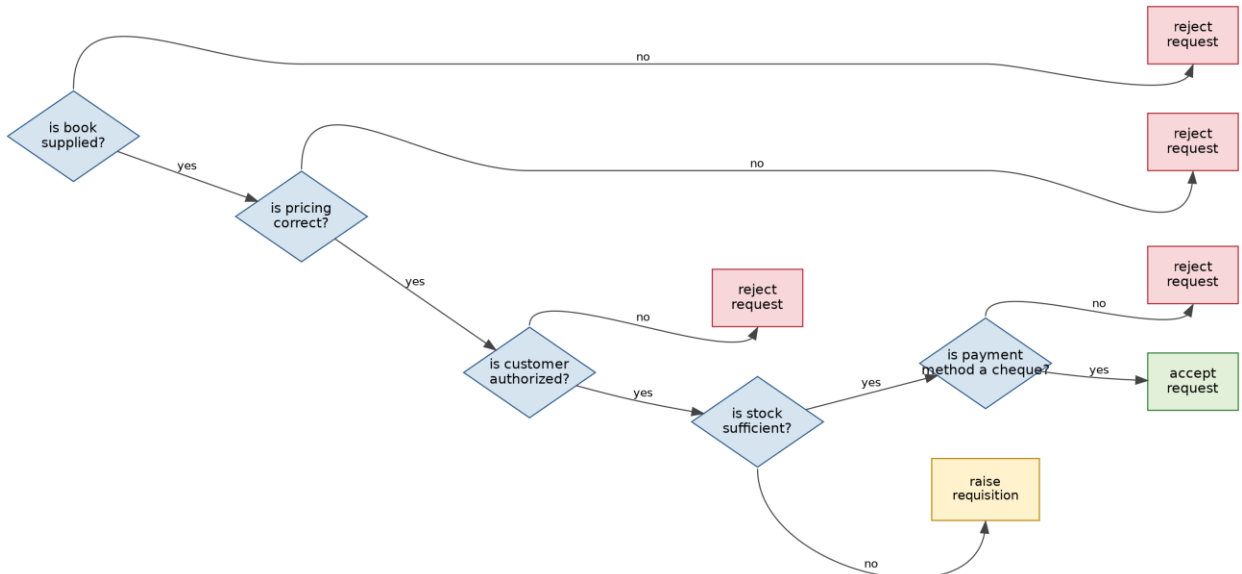


Figure 9 — Decision Tree for order acceptance (5 decisions)

11.2. Decision Table — Order Validation Business Rules

Requirement Number						
Condition	1	2	3	4	5	6
Book is supplied by publisher	F	T	T	T	T	T
Pricing is correct	—	F	T	T	T	T
Customer is authorized	—	—	F	T	T	T
Quantity on Hand is sufficient	—	—	—	F	T	T
Payment Method is Cheque	—	—	—	—	F	T
Action						
Accept order						X
Reject order	X	X	X		X	
Raise purchase requisition				X		

12. Event–Response Table

ID	Event	System State	System Response
1	Customer submits a sales order	No order in progress	Validate item availability and pricing against the catalogue
2	Order line fails validation	Validating order	Return order to customer with reason (item not supplied or pricing incorrect)
3	Order passes validation	Validated	Check Quantity on Hand for each order line
4	Order line stock is insufficient	Checking stock	Raise a purchase requisition (grouped by publisher); set status to Awaiting Stock
5	Goods Received Note recorded	Awaiting stock	Increase Quantity on Hand for the received ISBN(s); re-check orders awaiting that stock
6	Order fully in stock	Validated	Generate sales order acknowledgement and notify the warehouse; set status to Acknowledged
7	Customer submits payment method	Placing order	Verify payment method is a cheque; reject any other method
8	Picking ticket actioned by warehouse	Acknowledged	Enable dispatch note generation pre-filled from the picking ticket
9	Dispatch note created	Picking	Auto-draft the sales invoice from dispatch, order, and price data; set status to Dispatched
10	Invoice approved	Dispatched	Update the sales ledger, send invoice to customer, provide data to nominal ledger; set status to Invoiced
11	Cheque and remittance advice received	Invoiced	Present outstanding invoices for matching against the remittance advice
12	Cheque amount does not match remittance total	Matching payment	Flag discrepancy; block automatic invoice closure pending manual review
13	Cheque amount matches remittance total	Matching payment	Mark referenced invoices as Paid; update customer balance
14	End-of-day deposit requested	Payment recorded	Compile all same-day confirmed cheques into a bank deposit record for the cashier
15	Monthly statement cycle triggered	Any state	Generate and send a statement of outstanding invoices/balance to every customer with activity or a balance due

13. UI/UX Design Overview

The interface follows a clean, role-based dashboard style: a persistent left-hand navigation sidebar (grouped by feature area), a page header showing record status (e.g., a colored badge for Draft/Validated/Despatched/Paid), and a consistent action bar (Save, Validate, Print/Export, Cancel) on every form — matching the Dialogue Map above.

- Dashboard: role-specific summary cards (open orders, items awaiting stock, invoices due) plus quick-action buttons.
- New Sales Order form: header fields at top (customer, terms), an editable item-line grid below with live extended-price and order-total calculation, and inline validation messages next to any flagged line.
- Stock Availability view: a simple traffic-light indicator (green = in stock, amber = partial, red = shortfall) per order line, with a one-click 'Raise Requisition' action on shortfall lines.
- Invoice / Statement screens: a document-style layout deliberately similar to the paper Sales Invoice and Remittance Advice samples in the case study, to ease staff transition.
- User journey (customer): Browse/search catalogue → build order → review order summary → submit → receive acknowledgement → track status → receive invoice/statement.
- Design style: modern, minimal, high-contrast status badges, card-based dashboards, and a restrained color palette (navy/blue accents) consistent with a professional B2B ordering tool.

14. Technical Design

14.1. Suggested Technology Stack

Layer	Technology
Frontend	React (component-based UI, form validation, responsive dashboard views)
Backend / API	Node.js with Express (REST API) or an equivalent framework
Database	PostgreSQL (relational, transactional integrity for orders/invoices/payments)
Authentication	Role-based auth (JWT sessions) for staff; lightweight customer verification for order placement
Document generation	Server-side PDF generation for acknowledgements, invoices, and statements
Hosting	Cloud application hosting (e.g., a managed PaaS) with a managed database service

14.2. System Architecture

A three-tier, layered architecture: (1) Presentation tier — the React web client; (2) Application/business-logic tier — the REST API enforcing validation rules, stock rules, and workflow state transitions (see State Chart); (3) Data tier — the relational database. The Purchase Order/Publisher interface and the Nominal Ledger interface (SRS Section 4.3) are implemented as outbound integration modules at the application tier, kept decoupled from the core order-to-cash workflow so they can be upgraded independently in later releases (e.g., replacing the file/print publisher interface with a publisher portal API).

14.3. Database Design

Table	Key Fields
Books	ISBN (PK), Title, Publisher_ID (FK), List_Price, Quantity_On_Hand
Publishers	Publisher_ID (PK), Name, Address, Contact_Ref
Customers	Customer_ID (PK), Name, Billing_Address, Shipping_Address, Customer_Ref
Sales_Orders	Order_No (PK), Customer_ID (FK), Order_Date, Status, Shipping_Terms, Payment_Terms, Delivery_Terms
Sales_Order_Items	Order_No (FK), Line_No, ISBN (FK), Quantity, Price_Each, Extended_Price
Purchase_Requisitions	Requisition_No (PK), Publisher_ID (FK), Requested_By, Authorised_By, Requisition_Date
Purchase_Orders	PO_No (PK), Requisition_No (FK), Publisher_ID (FK), Order_Date, Delivery_Date
Goods_Received_Notes	GRN_No (PK), PO_No (FK), Date_Received, Carrier, Received_By
Picking_Tickets	Ticket_No (PK), Order_No (FK), Status
Despatch_Notes	Despatch_No (PK), Order_No (FK), Carrier, Despatch_Date
Invoices	Invoice_No (PK), Despatch_No (FK), Invoice_Date, Total_Amount, Status
Payments	Remittance_No (PK), Customer_ID (FK), Cheque_Amount, Date_Received
Payment_Invoice_Match	Remittance_No (FK), Invoice_No (FK), Amount_Applied

15. Testing Plan

15.1. Test Strategy

Testing combines unit tests (validation logic, price/total calculations, stock-check logic), integration tests (order-to-invoice workflow across features), and user-acceptance testing with sales, warehouse, and accounts staff using scenarios drawn directly from the case study's sample documents, to confirm generated documents match the expected fields and totals.

15.2. Test Cases

ID	Test Case	Steps	Expected Result
TC-01	Reject order for unsupplied book	Submit an order line with an ISBN not in the catalogue	Order line rejected; order returned to customer with reason
TC-02	Reject order for incorrect pricing	Submit an order line with a price differing from the catalogue price	Order line flagged as invalid pricing
TC-03	Accept order with sufficient stock	Submit a valid order where Quantity on Hand $\geq$ ordered quantity	Acknowledgement generated automatically; status = Acknowledged
TC-04	Raise requisition on shortfall	Submit a valid order exceeding Quantity on Hand	Purchase requisition created, grouped by publisher; status = Awaiting Stock
TC-05	Extended price / order total calculation	Enter multiple lines with quantity and price	Extended price = qty $\times$ price; order total = sum of extended prices
TC-06	Picking ticket excludes backordered items	Generate a picking ticket for an order with mixed availability	Only in-stock lines appear on the picking ticket
TC-07	Invoice auto-drafted from dispatch	Create a dispatch note for a dispatched order	Invoice is generated with matching items, quantities, and totals, with no manual re-entry
TC-08	Payment matching with discrepancy	Record a cheque amount that does not equal the remittance advice total	System flags discrepancy; invoices remain unpaid pending review
TC-09	Successful payment matching	Record a cheque matching the remittance advice total exactly	Referenced invoices marked Paid; customer balance reduced accordingly
TC-10	Role-based access enforcement	Attempt to mark an invoice paid while logged in as Sales Staff	Action denied; only Credit Control role permitted

15.3. Edge Cases

- An order containing a mix of valid and invalid lines (partial rejection handling).
- Two orders competing for the last remaining units of Quantity on Hand (concurrency).
- A customer with a zero or negative outstanding balance at statement time (should be excluded per the case study's rule of 'transaction occurred or balance outstanding').
- A dispatch note attempting to reference quantities greater than the picking ticket allows (should be blocked, per the no-partial-shipment rule).

15.4. Acceptance Criteria

- All High-priority functional requirements in SRS Section 3 have at least one passing test case.
- Generated acknowledgement, picking ticket, dispatch note, invoice, and statement documents reproduce the same fields and calculation logic shown in the case study's sample documents.
- No test case results in a partial shipment or a duplicate invoice/ledger update for the same event.

16. Risk Analysis

Risk	Likelihood	Impact	Mitigation
Incomplete or inconsistent data migration from manual records	Medium	High	Run a parallel data-reconciliation pass before cut-over; keep the manual system available during transition.
Scope creep beyond the Release 1.0 feature set	Medium	Medium	Enforce the Vision and Scope document's release boundaries; defer additions to Release 2.0/3.0.
Underestimated effort for diagram/documentation consistency across deliverables	Medium	Medium	Single shared glossary and traceability from feature tree → use cases → SRS requirements.
Publisher-side manual process becomes a bottleneck once internal steps are automated	Low	Medium	Keep Phase 1 requisition/PO output print/file-friendly; monitor for Release 3.0 portal justification.

17. Future Enhancements

- Customer self-service portal with full order history and tracking (Release 2.0, per Vision & Scope).
- Electronic payment gateway to reduce reliance on cheque-based remittance.
- Publisher-facing portal for electronic purchase orders and goods-received confirmation.
- Business-intelligence dashboards and demand forecasting for stock replenishment.
- Multi-currency support for a broader international customer base.

18. Assumptions Summary

- Existing Books, Customers, and Publishers data can be exported and migrated.
- Publisher and banking interfaces remain manual/paper-compatible in Release 1.0.
- The Sales System is the sole in-scope system; the nominal ledger is an existing external ledger receiving data from this system.
- All figures and currency are in EGP, matching the case study's sample documents.